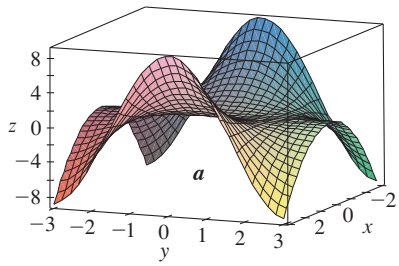
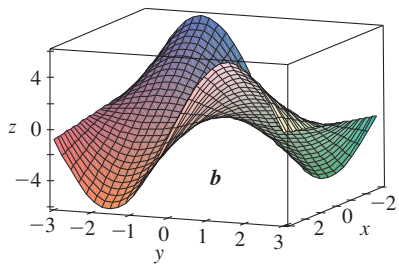
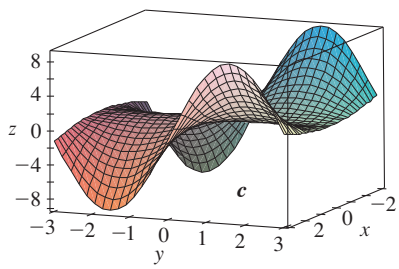
Suggested problems: 1-39 odd

Problem 1: The following surfaces, labeled a, b, and c, are graphs of a function f and its partial derivatives f_x and f_y . Identify each surface and give reasons for your choices.







Problem 2: The temperature at a point (x, y) on a flat metal plate is given by

$$T(x, y) = \frac{60}{1 + x^2 + y^2},$$

where T is measured in °C and x and y in meters. Find the rate of change of temperature with respect to distance at the point $(2, 1)$ in the x-direction and the y-direction.